

Elyse Rosenbaum

professor

基本信息

姓名: Elyse Rosenbaum

所属院校: 伊利诺伊大学厄巴纳-香槟

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地区: United States

年龄范围: 50岁以上

企业经历: 有

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可, 展现了卓越的研究能力和影响力。

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

模拟与射频IC设计

EDA工具与电路验证

微电子封装与可靠性

数字集成电路设计

模拟与数模混合电路

低功耗电路与能量管理

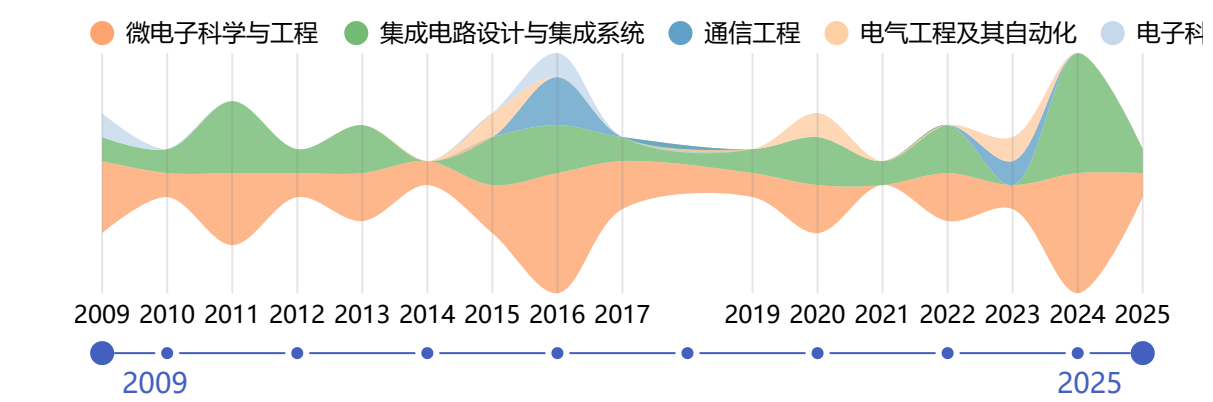
超大规模集成电路设计

个人简介

Elyse Rosenbaum is the Melvin and Anne Louise Hassebrock Professor in Electrical and Computer Engineering at the University of Illinois at Urbana - Champaign and the Acting Associate Dean for Research in the Grainger College of Engineering. She holds degrees from Cornell, Stanford, and UC Berkeley. She directs the CAEML. Her research interests include machine

learning in electronics and ESD - related topics. She has taught multiple ECE courses, authored over 200 papers, and served as an editor for IEEE journals. She has received numerous awards and is an IEEE Fellow. Contact: elyse AT illinois DOT edu, 1308 W. Main St, Urbana, IL 61801.

近年发表的论文



Exploiting Chiplet Integration Technology for Fast High-Capacity DRAM Modules
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS
Xia, Zihan; Song, Chihun; Krishna, Ram; Victor, Ashita; Penta, Srujan; Bakir, Muhannad S.; Rosenbaum, Elyse; Kim, Nam Sung; Kang, Ming
被引用次数: 0 2025

On-Chip ESD Protection for Multi-Gbps Automotive Serial IO in a 16-nm FinFET Process
2024 46TH ANNUAL EOS/ESD SYMPOSIUM, EOS/ESD 2024
Huang, Shudong; Parthasarathy, Srivatsan; Zhou, Yuanzhong (Paul); Hajjar, Jean-Jacques; Rosenbaum, Elyse
被引用次数: 0 2024

Reduced RC Time Constant High Voltage Tolerant Supply Clamp for ESD Protection in 16nm FinFET Technology
2024 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM, IRPS 2024
Huang, Shudong; Parthasarathy, Srivatsan; (Paul) Zhou, Yuanzhong; Hajjar, Jean-Jacques; Rosenbaum, Elyse
被引用次数: 0 2024

On-Chip Single-Shot Pulse Generator for TDDDB Characterization on a Sub-nanosecond Timescale
2024 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM, IRPS 2024
Drallmeier, Matthew; Zhou, Yujie; Rosenbaum, Elyse
被引用次数: 0 2024

Accuracy Preserving Extensions to a PDK MOSFET Model for ESD Simulation

2024 46TH ANNUAL EOS/ESD SYMPOSIUM, EOS/ESD 2024

Zhou, Yujie; Rosenbaum, Elyse

被引用次数: 0

2024

Compact Models for Simulation of On-Chip ESD Protection Networks

IEEE TRANSACTIONS ON ELECTRON DEVICES

Rosenbaum, Elyse; Huang, Shudong; Drallmeier, Matthew; Zhou, Yujie

被引用次数: 1

2024

Physics-based Compact Model of N-Well ESD Diodes

2023 45TH ANNUAL EOS/ESD SYMPOSIUM, EOS/ESD

Huang, Shudong; Rosenbaum, Elyse

被引用次数: 2

2023

Distributed Protection for High-Speed Wireline Receivers

2023 45TH ANNUAL EOS/ESD SYMPOSIUM, EOS/ESD

Drallmeier, Matthew; Rosenbaum, Elyse

被引用次数: 4

2023

A High Voltage Tolerant Supply Clamp for ESD Protection in a 45-nm SOI Technology

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Huang, Shudong; Parthasarathy, Srivatsan; Zhou, Yuanzhong (Paul); Hajjar, Jean-Jacques; Rosenbaum, Elyse

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A High Voltage Tolerant Supply Clamp for ESD Protection in a 45-nm SOI Technology

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Huang, Shudong; Rosenbaum, Elyse; Parthasarathy, Srivatsan; Zhou, Yuanzhong (Paul); Hajjar, Jean-Jacques

被引用次数: 1

2022

A Methodology to Optimize the Number and Placement of Decoupling Capacitors in a Multilevel Power Delivery Network

2022 IEEE ELECTRICAL DESIGN OF ADVANCED PACKAGING AND SYSTEMS (EDAPS)

Krishna, Ram; Nguyen, Thong; Watanabe, Atom O.; Becker, Dale; Kumar, Arvind; Rosenbaum, Elyse

被引用次数: 1

2022

Neural Ordinary Differential Equation Models of Circuits: Capabilities and Pitfalls

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Xiong, Jie; Yang, Alan; Raginsky, Maxim; Rosenbaum, Elyse

被引用次数: 1

2022

Statistical Learning of IC Models for System-Level ESD Simulation

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Xiong, Jie; Chen, Zaichen; Raginsky, Maxim; Rosenbaum, Elyse

被引用次数: 3

2021

Analysis and Design of Integrated Voltage Regulators for Supply Noise Rejection During System-Level ESD

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Xiu, Yang; Rosenbaum, Elyse

被引用次数: 1

2020

Numerical Methods for Event-Detection State Vector Simulation of Switched-Mode Power Supplies

IEEE JOURNAL OF EMERGING AND SELECTED TOPICS IN POWER ELECTRONICS

Reiman, Collin; Rosenbaum, Elyse

被引用次数: 0

2020

Analysis of System-Level ESD-Induced Soft Failures in a CMOS Microcontroller

IEEE TRANSACTIONS ON ELECTROMAGNETIC COMPATIBILITY

Vora, Sandeep; Rosenbaum, Elyse

被引用次数: 1

2020

Special Issue on Reliability

IEEE TRANSACTIONS ON ELECTRON DEVICES

Mahapatra, Souvik; Kerber, Andreas; Compagnoni, Christian Monzio; Koval, Randy; Meneghesso, Gaudenzio; Sheridan, David; Ramey, Stephen; Wang, Runsheng; Stathis, Jim; Chen, Kevin J.; Kaczer, Ben; Pancheri, Lucio; Rosenbaum, Elyse; Mouli, Chandra; Wong, Hei

被引用次数: 0

2019

Measurement and Simulation of On-Chip Supply Noise Induced by System-Level ESD

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Xiu, Yang; Thomson, Nicholas; Rosenbaum, Elyse

被引用次数: 4

2019

ESD Self-Protection of High-Speed Transceivers Using Adaptive Active Bias Conditioning

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Keel, Min-Sun; Rosenbaum, Elyse

被引用次数: 2

2017

Soft-Failures Induced by System-Level ESD

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Thomson, Nicholas A.; Xiu, Yang; Rosenbaum, Elyse

被引用次数: 25

2017

Full-Component Modeling and Simulation of Charged Device Model ESD

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Meng, Kuo-Hsuan; Shukla, Vrashank; Rosenbaum, Elyse

被引用次数: 7

2016

Compact distributed multi-finger MOSFET model for circuit-level ESD simulation

MICROELECTRONICS RELIABILITY

Meng, Kuo-hsuan; Chen, Zaichen; Rosenbaum, Elyse

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2016

A Study of BER-Optimal ADC-Based Receiver for Serial Links

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Lin, Yingyan; Keel, Min-Sun; Faust, Adam; Xu, Aolin; Shanbhag, Naresh R.; Rosenbaum, Elyse; Singer, Andrew C.

被引用次数: 17

2016

Physical Basis for CMOS SCR Compact Models

IEEE TRANSACTIONS ON ELECTRON DEVICES

Mertens, Robert; Rosenbaum, Elyse

被引用次数: 8

2016

CDM-Reliable T-Coil Techniques for a 25-Gb/s Wireline Receiver Front-End

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Keel, Min-Sun; Rosenbaum, Elyse

被引用次数: 10

2016

Charged Device Model Reliability of Three-Dimensional Integrated Circuits

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Shukla, Vrashank; Rosenbaum, Elyse

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2015

Analysis of Active-Clamp Response to Power-On ESD: Power Supply Integrity and Performance Tradeoffs

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Mertens, Robert; Thomson, Nicholas; Xiu, Yang; Rosenbaum, Elyse

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Piecewise-Linear Model With Transient Relaxation for Circuit-Level ESD Simulation

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Meng, Kuo-Hsuan; Mertens, Robert; Rosenbaum, Elyse

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Prediction of Charged Device Model Peak Discharge Current for Microelectronic Components

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Shukla, Vrashank; Boselli, Gianluca; Dissegna, Mariano; Duvvury, Charvaka; Sankaralingam, Raj; Rosenbaum, Elyse

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2014

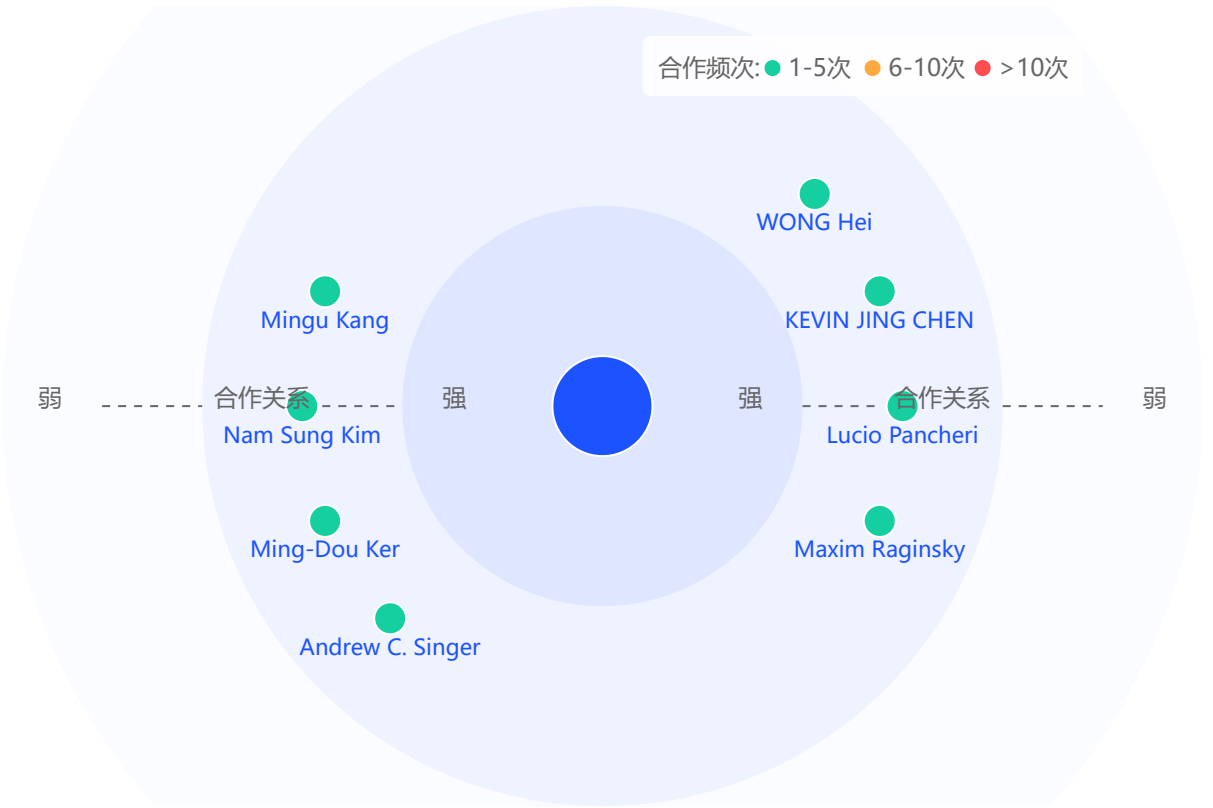
Verification of Snapback Model by Transient I-V Measurement for Circuit Simulation of ESD Response

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被引用次数: 8	2013
Comparison of FICDM and Wafer-Level CDM Test Methods	
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Diode-Triggered Silicon-Controlled Rectifier With Reduced Voltage Overshoot for CDM ESD Protection	
IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY	
Chen, Wen-Yi; Rosenbaum, Elyse; Ker, Ming-Dou	
被引用次数: 46	2012
Modeling and Understanding of External Latchup in CMOS Technologies-Part II: Minority Carrier Collection Efficiency	
IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY	
Farbiz, Farzan; Rosenbaum, Elyse	
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Modeling and Understanding of External Latchup in CMOS Technologies-Part I: Modeling Latchup Trigger Current	
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Comparison of Wafer-Level With Package-Level CDM Stress Facilitated by Real-Time Probing	
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Jack, Nathan; Shukla, Vrashank; Rosenbaum, Elyse	
被引用次数: 4	2011
A Scalable SCR Compact Model for ESD Circuit Simulation	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Di Sarro, James P; Rosenbaum, Elyse	
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A new compact model for external latchup	
MICROELECTRONICS RELIABILITY	
Farbiz, Farzan; Rosenbaum, Elyse	
被引用次数: 1	2009
Layout Optimization of ESD Protection Diodes for High-Frequency I/Os	
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Bhatia, Karan; Jack, Nathan; Rosenbaum, Elyse	
被引用次数: 26	2009
Oscillatory Transmission Line Pulsing for Characterization of Device Transient Response	

学术合作



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